AEGIS

Autonomous Commerce Operating System

PROOF OF CONCEPT

AI proposes. Policy authorizes. Razorpay executes.

Project by

Sanjeev Kumar S

sanjeevkumaredu8@gmail.com

Razorpay Buildathon · Track 01 — AI Growth & Agentic Commerce

1. Executive Summary

AEGIS is an AI-native commerce operating system designed to make merchants transactable by autonomous AI buyers while keeping every money action explainable, bounded, and gated. The system separates probabilistic commerce intelligence from a deterministic trust boundary that decides whether a proposed transaction is actually authorized.

The proof of concept demonstrates an end-to-end flow: natural-language shopping intent is normalized into structured constraints, candidate products are ranked, an AI purchase proposal is generated, authorization is cryptographically bound to the approved intent, adversarial mutations are intercepted, and only an authorized transaction is allowed to reach Razorpay Test Mode.

2. Problem Statement

Autonomous agents can discover products, reason over offers, and initiate actions, but traditional payment flows were designed around a human explicitly approving each transaction. This creates a critical trust gap between what an AI agent proposes and what a payment system ultimately executes.

Key risks include amount escalation, prompt injection, poisoned catalog data, recipient substitution, category substitution, and tampering with transaction parameters. AEGIS addresses this gap by inserting a deterministic, auditable authorization boundary between agentic decision-making and payment execution.

3. Track 01 Alignment

Track objective	**AEGIS implementation**
Grow merchant revenue	AI-driven discovery and candidate ranking improve product visibility and selection-fit for machine-mediated demand.
Make merchants sellable to AI buyers	Structured, AI-readable catalog data exposes product attributes, availability, merchant trust, fulfillment and discoverability signals.
Agentic commerce	Natural-language intent → discovery → selection → authorization → Razorpay Test Mode checkout.
Bar: explainable, bounded, gated	Every purchase proposal exposes decision factors; hard constraints gate execution; failed attacks are blocked with evidence recorded.

4. Proposed Solution

AEGIS introduces a two-plane architecture. The Probabilistic Commerce Plane handles intent understanding, discovery, graph-based representation, ranking and optimization. The Deterministic Trust Boundary handles authorization, policy enforcement, runtime interception and payment gating.

Architectural invariant: NO VERIFICATION = NO RAZORPAY ORDER.

5. System Architecture

Probabilistic Commerce Plane

• Intent understanding and structured constraint extraction.
• MiniLM-based semantic representation for intent matching.

• Heterogeneous commerce graph connecting products, merchants, categories and attributes.
• GraphSAGE-based candidate representation and ranking.
• Selection-fit scoring and counterfactual optimization for growth-oriented recommendations.

Deterministic Trust Boundary

• HMAC-SHA256 Root Intent Certificate binds the approved intent to the authorization context.
• Policy invariants enforce budget, merchant, category and derivation continuity.
• Runtime interception validates proposed actions before payment execution.
• Fail-closed payment gate suppresses unauthorized Razorpay order creation.
• Append-only SHA-256 hash-chained ledger preserves an auditable evidence trail.

6. AI Growth & Agentic Commerce

01	AI-Readable Commerce Catalog	Structures merchant inventory for AI discovery, comparison and transaction.
02	Autonomous Product Discovery	Maps natural-language intent to relevant commerce candidates.
03	AI Growth & Ranking Engine	Ranks candidates using intent fit, quality, trust, availability and fulfillment signals.
04	Merchant AI Discoverability	Makes catalog information more machine-readable and discoverable to AI buyers.
05	Conversational Agentic Checkout	Connects discovery, selection, authorization and payment into one flow.
06	Explainable Purchase Decisions	Exposes selection score and decision factors before money moves.
07	Bounded & Gated Transactions	Enforces budget, merchant, category and authorization constraints.
08	Failure-Safe Payment & Audit	Blocks invalid actions, keeps money movement at zero, and records evidence.

7. End-to-End Proof of Concept Flow

Example intent: “Hotels in Goa under ₹3,000.”

AEGIS converts the request into structured constraints: category = hotel, location = Goa, maximum budget = ₹3,000. Six candidates are evaluated using structured product and merchant signals. The ranking layer produces a selection-fit score rather than simply choosing the cheapest option.

The selected proposal is then passed to the trust boundary with the exact product identity, merchant, amount, intent and relevant attributes. The trust boundary validates the proposal continuously before permitting payment execution.

8. Security & Attack Lab

Amount Escalation	Unauthorized increase in transaction value beyond the approved budget.
Poisoned Catalog	Manipulated product data attempts to influence the AI selection.
Recipient Substitution	Payment is redirected to an unapproved merchant or recipient.
Category Substitution	The approved product category is changed during execution.
Derivation Tampering	Authorized transaction parameters are altered after approval.
Prompt Injection	Malicious instructions attempt to override the original purchase intent.

Demonstrated failure behavior: Attack detected → authorization denied → Razorpay order suppressed → ₹0 moved → evidence recorded.

9. Authorization Model

AEGIS asks four deterministic questions before a money action is allowed:

• Is the amount within the authorized ceiling?
• Is the merchant authorized?
• Is the category within scope?
• Is the derivation state valid?

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10. Auditability & Evidence

Every relevant authorization and transaction state is persisted in an append-only, cryptographically linked audit ledger. The ledger supports verification of state continuity and provides a forensic trail for both successful transactions and blocked attacks.

The evidence layer is designed around three outcomes: what the agent proposed, why authorization was granted or denied, and whether money was actually allowed to move.

11. Demo / CLI Runbook

```
python backend/cli/main.py system status

python backend/cli/main.py agent run --intent "hotel in Goa under 3000"

python backend/cli/main.py growth predict --query "hotel in Goa under 3000"

python backend/cli/main.py growth counterfactual

python backend/cli/main.py attack run amount-escalation
```

```
python backend/cli/main.py ledger tail -n 5

python backend/cli/main.py ledger verify
```

12. Expected POC Outcomes

• AI buyer can discover and rank merchant offers from natural-language intent.
• Merchant products expose structured signals useful for machine-mediated commerce.
• Every payment proposal is bounded by explicit authorization constraints.
• Adversarial mutations are intercepted before financial execution.
• Successful payments can reach Razorpay Test Mode only after verification.
• Blocked and successful actions leave a verifiable audit trail.

13. Conclusion

AEGIS turns autonomous commerce from an unrestricted agent problem into a governed execution system. It combines AI-native commerce growth with a deterministic trust boundary so merchants can become sellable to AI buyers without giving autonomous agents unrestricted authority over money.

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